

Hailin Wang

Ph.D. candidate at Tongji University & The Hong Kong Polytechnic University

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About me

I am Hailin Wang, a passionate researcher and developer in the field of geotechnical engineering. My work focuses on the application of multiscale micromechanical methods to understand the behavior of geomaterials under various loading conditions. I have a strong interest in computational modeling, numerical simulations, and the integration of machine learning techniques into geotechnical analysis.

Education

BEng	Huazhong University of Science and Technology , Civil Engineering	Wuhan, China
	- Weighted grade: 91.79 (rank 1/64) - Honors: Received the National Scholarship and Merit Student of Huazhong University of Science and Technology for three consecutive years	Sept 2016 – June 2020
PhD	Tongji University , Geotechnical Engineering	Shanghai, China
	Advisor: Prof. Xiaoqiang Gu	Sept 2020 – present
PhD	The Hong Kong Polytechnic University , Geotechnical Engineering	Hong Kong, China
	- Thesis: Multiscale Modelling of Granular Soils with Inherent and Induced Fabric Anisotropy - Advisor: Prof. Zhen-Yu Yin	Sept 2021 – present

Projects

abqpy	June 2022 – present
Type Hints for Abaqus/Python Scripting	
- Enable intelligent type hinting for Abaqus/Python scripting 300+ GitHub stars, 50+ forks	

Publications

Uncertainty Quantification of Soil-Structure Interface Properties with an Enhanced Hypoplastic Interface Model	June 2024
Hai-Lin Wang , Yin-Fu Jin, Zhen-Yu Yin, Xiao-Qiang Gu (International Journal of Geomechanics)	
Evaluation of Soil-Structure Interface Models	July 2024
Hai-Lin Wang , Zhen-Yu Yin, Yin-Fu Jin, Xiao-Qiang Gu 10.1061/JGGEFK.GTENG-11486 (Journal of Geotechnical and Geoenvironmental Engineering)	
Evaluation of Soil-Structure Interface Models Considering Cyclic Loading Effect	Nov 2024
Hai-Lin Wang , Zhen-Yu Yin, Xiao-Qiang Gu, Yin-Fu Jin 10.1002/nag.3831 (International Journal for Numerical and Analytical Methods in Geomechanics)	

Integration of critical state theory into a micromechanical model for granular materials accounting for fabric evolution Hai-Lin Wang , Xiaoqiang Gu, Zhen-Yu Yin, Chao-Fa Zhao 10.1016/j.compgeo.2025.107379 (Computers and Geotechnics)	Oct 2025
A generalized anisotropic failure criterion for granular materials based on a multi-scale micromechanical approach Hai-Lin Wang , Xiaoqiang Gu, Zhen-Yu Yin 10.1016/j.ijsolstr.2025.113696 (International Journal of Solids and Structures)	Jan 2026
Micromechanics-inspired modeling of granular materials Hai-Lin Wang , Xiaoqiang Gu, Zhen-Yu Yin 10.1016/j.ijsolstr.2026.114263 (International Journal of Solids and Structures)	Oct 2026

Selected Honors

- National Scholarship (2017-2019)
- Merit Student of Huazhong University of Science and Technology (2017-2019)

Skills

Research Areas: Geotechnics, Geomechanics

Languages: Python, C++, Matlab, Fortran